

Amogh Jayasimha

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Education

Northeastern University

Boston, MA

Master of Science in Information Systems, **GPA: 3.84**

Expected: Aug 2026

Relevant Courses: CSYE 6225 – Cloud Computing, TELE 6420 – Infrastructure Automation Design and Tools (Puppet, Ansible)

PES University

Bangalore, India

Bachelor of Technology in Electronics and Communication Engineering

Graduation Date: June 2022

Technical Skills

- **Languages and technologies:** Python (OOP, FastAPI, Flask, Boto3), Golang, Bash, PowerShell, JavaScript, Git, PostgreSQL, MySQL, Redis, Streamlit, Rust (learning); Linux/UNIX
- **Cloud & Platforms:** AWS (EC2, IAM, KMS, CloudWatch), GCP (GKE, Compute Engine, IAM, Ops Suite), OpenStack
- **Containers & Kubernetes:** Docker, Kubernetes, Helm, ArgoCD, Kargo, KEDA, Kubernetes NetworkPolicies, cert-manager, Chaos Mesh, Operators, Stateful services
- **CI/CD, GitOps & Security:** GitHub Actions, GitOps promotion pipelines, CI/CD optimization, Terraform, Cosign/Sigstore, Kyverno, External Secrets Operator
- **Observability & Reliability:** Prometheus, Grafana, Loki, CloudWatch, incident response and production debugging

Work Experience

Radius (EVM-based Blockchain Infrastructure)

Cambridge, MA

Platform / Infrastructure Engineering Co-op

Sep 2025 – Dec 2025

- Optimized PR deployment workflows by differentiating config-only versus code changes and using registry-level image retagging, reducing deployment time from ~17–27 minutes to ~2 minutes while preserving isolation and correctness.
- Designed and implemented default-deny Kubernetes NetworkPolicies for EVM-based blockchain services, enforcing component-scoped ingress/egress rules, DNS allowlists, and explicit infrastructure access to harden workloads before production rollout.
- Responded to chaos-testing incidents by adding persistent volumes and anti-affinity rules for stateful services (storage layers, shard managers), preventing data loss and cascading failures during node rotation.
- Unified Kargo promotion pipeline (benchmarking → staging → testnet → mainnet) with native GitHub Deployments and ArgoCD notifications for end-to-end deployment traceability with Slack alerts and compare URLs.
- Contributed to operator maturity by aligning CI and Helm workflows with operator defaults and validation logic, reducing reliance on brittle CRD-required fields and improving safety of operator-managed deployments.

Societe Generale

Bangalore, India

Systems DevOps Engineer

July 2022 – July 2024

- Automated 200+ legacy server migrations using shell scripts and Python FastAPI microservices, reducing manual effort by 40%.
- Built GitOps pipelines with GitHub Actions to auto-provision and deploy IaaS systems, reducing provisioning time from 2 hrs to 25 min; integrated secrets and monitoring for DevOps resiliency.
- Developed Python/Airflow pipelines to enforce PCI-DSS compliance checks and orchestrate parallel post-migration validation workflows across cloud perimeters.
- Reduced cloud costs by 25% via right-sizing automation (Python/Boto3), earning Spot Award from Global Head.
- Created observability stack (Prometheus/Grafana) for 150+ cloud instances across AWS and GCP environments.

Open Source

Kubernetes Supply Chain Security Platform

[GitHub](#)

Kyverno admission control enforcing cryptographic image signature verification (Cosign/Sigstore), SBOM validation, and CVE-blocking policies before containers reach production. Featured in the official Kyverno community showcase as a reference implementation of supply chain security.

Projects

k8s-chaos-promotion: Chaos-Gated GitOps Promotion Pipeline

[GitHub](#) — *In Development*

- GitOps pipeline on GKE where staging cannot promote to production unless the app survives live Chaos Mesh fault injection (pod kills, network latency, RPC failures) under k6 load; chaos suite runs inside a Kargo AnalysisTemplate verification gate – a pattern with no prior public reference implementation.
- Provisioned GKE, VPC-native networking, and GCS state via Terraform with fully keyless architecture (Workload Identity Federation, GitHub Actions OIDC); wired ArgoCD App-of-Apps, Kargo promotion, ESO, and Prometheus/Grafana/Loki for full observability.
- Integrated x402-style HTTP micropayment flow with on-chain settlement via Radius EVM (web3.py, Foundry); validated with ~132 real on-chain transactions in testnet across dev and staging.

Fully Automated Cloud-Native FastAPI Application on AWS & GCP

[GitHub](#) — Jan 2025 – Apr 2025

- Built shared Packer templates for AWS EC2 and GCP Compute Engine; automated multi-cloud deployment pipelines via GitHub Actions (OIDC) and Terraform with PostgreSQL, KMS, Secrets Manager, and Auto Scaling.
- Configured CloudWatch & GCP Ops Suite monitoring; enforced IAM least-privilege policies and KMS encryption for data at rest across both cloud environments.